

# MDS211 Nervous System — SA1 Mock Exam 2

## Lectures 1–16 · 77 Questions · Single Best Answer

This is a SECOND, fully independent 77-question set — zero overlap with Mock Exam 1 — using the same real SA1 weighting from your official schedule (proportional to actual lecture hours, 15 hours total across Lec 1–16). Every question again comes from my verified, fact-checked bank (the same one behind “The Professor’s Gauntlet”), not the error-prone original archive. Use this as a second timed practice run once you’ve reviewed Mock Exam 1’s answer key — budget roughly 2 minutes per question, matching the real SA1’s pacing.

### Lecture 1 — Introduction & Organization of the CNS (5 questions)

1. Which statement about gray and white matter distribution is correct?
  - A. In the spinal cord, gray matter is peripheral and white matter is central
  - B. In the cerebrum, gray matter forms the cortex and white matter lies deep to it
  - C. The cerebellum has no gray matter
  - D. White matter contains neuron cell bodies while gray matter contains only axons
  - E. Gray and white matter are distributed identically in brain and spinal cord

*† original question, written from lecture content*
2. A newborn is found to have a portion of spinal cord and meninges protruding through a vertebral defect, covered only by a thin membrane. This is best classified as:
  - A. Spina bifida occulta
  - B. Meningocele
  - C. Myelomeningocele
  - D. Anencephaly
  - E. Encephalocele

*† original question, written from lecture content*
3. Which cranial nerve numbers correspond to nerves that are purely sensory?
  - A. III, IV, VI
  - B. I, II, VIII
  - C. V, VII, X
  - D. XI, XII
  - E. III, VII, X

*† original question, written from lecture content*
4. Dorsal root ganglia contain the cell bodies of which type of neuron?
  - A. Lower motor neurons
  - B. Preganglionic autonomic neurons
  - C. First-order (primary) sensory neurons
  - D. Interneurons of the dorsal horn
  - E. Postganglionic sympathetic neurons

*† original question, written from lecture content*

5. A patient has a lesion that damages both anterior and posterior spinal nerve roots equally at one level. Which functional deficit pattern is expected?

- A. Pure sensory loss only
- B. Pure motor loss only
- C. Combined sensory and motor loss at that segmental level
- D. No deficit, since roots regenerate quickly
- E. Only autonomic dysfunction

*† original question, written from lecture content*

## Lecture 2 — Neurohistology (Neuron & Neuroglia) (3 questions)

6. If the tract carrying FAST, sharp, well-localized pain is damaged, which fiber type is responsible for that lost sensation?

- A. A- $\beta$  fiber
- B. C fiber (unmyelinated)
- C. Paleospinothalamic tract fibers
- D. Neo-spinothalamic tract (A- $\delta$ ) fibers
- E. A- $\alpha$  fiber

7. Which of the following is NOT a component of the blood-brain barrier?

- A. Oligodendrocyte
- B. Astrocyte end-feet
- C. Pericyte of capillary
- D. Endothelial cell tight junctions
- E. Basement membrane

8. Which cell type is the resident immune cell of the CNS, analogous to macrophages elsewhere in the body?

- A. Microglia
- B. Schwann cell
- C. Oligodendrocyte
- D. Astrocyte
- E. Ependymal cell

## Lecture 3 — Properties of Neuron: RMP, Action Potential, Conduction (5 questions)

9. Which of the following correctly describes when Na<sup>+</sup>/K<sup>+</sup>-ATPase activity increases the most?

- A. During periods of high neuronal firing, to restore ionic gradients dissipated by Na<sup>+</sup> influx/K<sup>+</sup> efflux
- B. Only during sleep
- C. Only in glial cells, never in neurons
- D. It runs at a constant, unchanging rate regardless of activity
- E. Only during hyperpolarization

**10.** When depolarizations from multiple simultaneous synaptic inputs on different parts of a neuron add together, this is called:

- A. Temporal summation
- B. Spatial summation
- C. Facilitation
- D. Long-term potentiation
- E. Accommodation

**11.** All of the following are true of chemical synapses EXCEPT:

- A. They involve a synaptic delay of roughly 0.5-1 ms
- B. Transmission is unidirectional (pre- to postsynaptic)
- C. They can amplify a small presynaptic signal into a larger postsynaptic response
- D. Transmission is instantaneous with essentially zero delay
- E. Neurotransmitter release is  $\text{Ca}^{2+}$ -dependent

**12.** Which statement about graded potentials is INCORRECT?

- A. They decay in amplitude with distance (decremental conduction)
- B. Their amplitude is proportional to stimulus strength
- C. They can summate
- D. They obey the all-or-none law like action potentials
- E. They are typically local, subthreshold events

**13.** During the absolute refractory period, why can no new action potential be generated no matter how strong the stimulus?

- A. All voltage-gated  $\text{K}^+$  channels are closed
- B. Voltage-gated  $\text{Na}^+$  channels are inactivated and cannot reopen until the membrane repolarizes
- C. The membrane is hyperpolarized below threshold, requiring only a slightly stronger stimulus
- D.  $\text{Cl}^-$  channels are maximally open
- E. The neuron has run out of neurotransmitter

*† original question, written from lecture content*

## Lecture 4 — Metabolic Pathways of Neurons & Nervous Tissue (5 questions)

**14.** Why can hyperammonemia impair neuronal function even though ammonia detoxification happens mainly in astrocytes rather than neurons?

- A. Because astrocyte dysfunction/swelling disrupts the glutamate-glutamine cycle and lactate shuttle that neurons depend on for neurotransmitter recycling and fuel
- B. Because ammonia only affects neurons and never astrocytes
- C. Because neurons contain glutamine synthetase too, so they are equally affected
- D. Ammonia directly destroys myelin
- E. There is no real connection between astrocyte and neuron function

*† original question, written from lecture content*

**15.** Which transporter class is primarily responsible for astrocytic uptake of the lactate that astrocytes then export to neurons?

- A. GLUT3
- B. EAAT-2
- C. Monocarboxylate transporters (MCT1/4)
- D. Na<sup>+</sup>/K<sup>+</sup>-ATPase
- E. SNAT

*† original question, written from lecture content*

**16.** Which glucose transporter is characteristic of astrocytes, versus the one preferentially expressed in neurons?

- A. Astrocyte: GLUT1; Neuron: GLUT3
- B. Astrocyte: GLUT3; Neuron: GLUT1
- C. Both use GLUT4 exclusively
- D. Astrocyte: GLUT2; Neuron: GLUT5
- E. Neither cell type expresses glucose transporters

*† original question, written from lecture content*

**17.** The pentose phosphate pathway (PPP), activated in neurons especially under high glucose availability, primarily generates which product important for antioxidant defense?

- A. ATP directly
- B. NADPH, which regenerates reduced glutathione (GSH) to neutralize reactive oxygen species
- C. Lactate
- D. Glycogen
- E. Acetyl-CoA

*† original question, written from lecture content*

**18.** Which micronutrient deficiency is classically associated with Wernicke-Korsakoff syndrome, disrupting neuronal energy metabolism?

- A. Vitamin B1 (thiamine)
- B. Vitamin C
- C. Vitamin B12 only
- D. Folate only
- E. Vitamin D

*† original question, written from lecture content*

### **Lectures 5–6 — Embryonic Development of the Nervous System / Development of the Eye (5 questions)**

**19.** Which embryonic structure gives rise to the pineal gland?

- A. Roof of the diencephalon (epithalamus)
- B. Floor of the mesencephalon
- C. Telencephalon
- D. Roof of the myelencephalon
- E. Neural crest

20. Which functional component is characteristic of neurons in the hypoglossal nucleus, located at the level of the myelencephalon (medulla)?
- A. GSA (general somatic afferent)
  - B. SVA (special visceral afferent)
  - C. GVA (general visceral afferent)
  - D. SVE (special visceral efferent)
  - E. GSE (general somatic efferent)
21. Which structure does the optic cup differentiate into?
- A. Lens
  - B. Cornea epithelium
  - C. Retina (both neural and pigmented layers) and epithelium of the iris/ciliary body
  - D. Vitreous humor only
  - E. Extraocular muscles
22. Failure of the choroid (retinal) fissure to fuse normally results in which congenital anomaly?
- A. Coloboma (a gap/cleft in the iris, retina, or optic disc)
  - B. Cataract
  - C. Anophthalmia
  - D. Strabismus
  - E. Glaucoma
23. The spinal trigeminal nucleus (carrying facial pain/temperature) is derived from which embryonic structure?
- A. Telencephalon
  - B. Alar plate of the developing brainstem (sensory column)
  - C. Basal plate of the developing brainstem (motor column)
  - D. Neural crest exclusively
  - E. Notochord

### Lecture 7 — Spinal Cord & Blood Supply (5 questions)

24. A 53-year-old man is hospitalized for acute spinal cord infarction (anterior spinal artery territory). Which sensory modality is characteristically SPARED?
- A. Pain
  - B. Temperature
  - C. Vibration and proprioception (dorsal column)
  - D. Crude touch
  - E. Motor function
25. Which of the following is NOT part of the central nervous system?
- A. Spinal cord
  - B. Cerebellum
  - C. Dorsal root ganglion
  - D. Brainstem
  - E. Cerebral cortex

**26.** If the anterior spinal artery is occluded, which structure/function is spared, consistent with its blood supply territory?

- A. Corticospinal tract function
- B. Pain and temperature sensation
- C. Dorsal column (vibration/proprioception) function, supplied by the posterior spinal arteries instead
- D. Anterior horn motor neurons
- E. Lateral spinothalamic tract

**27.** Which finding is a primary MOTOR symptom of a complete spinal cord lesion (transection) below the level of injury?

- A. Loss of pain sensation
- B. Spastic paralysis (initially flaccid in spinal shock, then spastic) below the lesion
- C. Loss of vibration sense
- D. Autonomic dysreflexia exclusively
- E. Loss of smell

**28.** Which structure extends from the pia mater to the dura mater along the lateral aspect of the spinal cord, anchoring it and separating dorsal (posterior) from ventral (anterior) spinal nerve roots?

- A. Denticulate ligament
- B. Filum terminale
- C. Ligamentum flavum
- D. Arachnoid trabeculae
- E. Falx cerebri

### Lecture 8 — PNS: Cranial & Spinal Nerves, Nerve Plexuses (3 questions)

**29.** Which functional component is NOT found in the facial nerve (CN VII)?

- A. SVE (motor to muscles of facial expression)
- B. GVE (parasympathetic to lacrimal/salivary glands)
- C. SVA (taste, anterior 2/3 tongue)
- D. GSA (general sensation from a small area of the external ear)
- E. GVA (general visceral afferent from thoracic/abdominal viscera)

**30.** Which spinal nerves are NOT included in a somatic nerve plexus?

- A. C1-C4 (cervical plexus)
- B. T2-T12 (thoracic nerves, which mostly run as individual intercostal nerves)
- C. C5-T1 (brachial plexus)
- D. L1-S4 (lumbosacral plexus)
- E. All spinal nerves form a plexus

**31.** A 41-year-old woman has recurring episodes of sharp, electric-shock-like facial pain triggered by chewing and touching her cheek. Which nerve is most likely involved?

- A. Facial nerve (Bell's palsy)
- B. Trigeminal nerve (trigeminal neuralgia)
- C. Glossopharyngeal nerve
- D. Vagus nerve
- E. Accessory nerve

## Lectures 9–10 — Brainstem: External & Internal Structures / Reticular Formation (10 questions)

- 32.** Which tract's second-order neurons cross the midline at the level of the medulla oblongata (rather than immediately upon entering the spinal cord)?
- A. Lateral spinothalamic tract (crosses within 1-2 spinal segments)
  - B. Dorsal column-medial lemniscus pathway (crosses as internal arcuate fibers in the caudal medulla)
  - C. Anterior spinothalamic tract
  - D. Corticospinal tract (crosses at the medulla too, but as a first-order/UMN axon, not a sensory 2nd-order neuron)
  - E. None of the sensory pathways cross in the medulla
- 33.** Damage to which major component of the Ascending Reticular Activating System (ARAS) most directly impairs consciousness?
- A. Cerebellum
  - B. Spinal cord below C4
  - C. Peripheral nerves
  - D. Pons and midbrain (rostral brainstem reticular formation)
  - E. Basal ganglia exclusively
- 34.** What is the major anatomical site whose bilateral damage produces coma via ARAS disruption?
- A. Cerebral cortex alone, unilaterally
  - B. Pons and midbrain reticular formation
  - C. Cerebellar vermis
  - D. Spinal cord
  - E. Peripheral autonomic ganglia
- 35.** A patient presents with altered consciousness, irregular breathing patterns, and decorticate/decerebrate posturing. This clinical picture most directly suggests dysfunction at which level?
- A. Peripheral nerve only
  - B. Brainstem (reticular formation) dysfunction, possibly from herniation or direct injury
  - C. Isolated muscle disease
  - D. Isolated dorsal root ganglion pathology
  - E. Isolated retinal disease
- 36.** Which cranial nerve nuclei are found at the level of the pons (not the medulla or midbrain)?
- A. CN III, IV
  - B. CN V, VI, VII, VIII
  - C. CN IX, X, XI, XII
  - D. CN I, II
  - E. None; the pons has no cranial nerve nuclei
- † original question, written from lecture content*
- 37.** Which structure, if damaged bilaterally in the pons, classically produces 'locked-in syndrome' (preserved consciousness and vertical eye movement, but quadriplegia and inability to speak)?
- A. Bilateral ventral pons (corticospinal and corticobulbar tracts), sparing the tegmentum/reticular formation
  - B. Bilateral cerebral cortex
  - C. Bilateral cerebellum
  - D. Bilateral thalamus
  - E. Bilateral occipital lobe
- † original question, written from lecture content*

**38.** Why does a lesion of the medial longitudinal fasciculus (MLF) classically cause internuclear ophthalmoplegia (INO)?

- A. The MLF connects the abducens nucleus on one side to the contralateral oculomotor (medial rectus) subnucleus, coordinating conjugate horizontal gaze
- B. The MLF is a purely sensory tract with no motor function
- C. The MLF controls only vertical gaze
- D. The MLF has no connection to cranial nerve nuclei
- E. The MLF is located in the spinal cord, unrelated to eye movement

*† original question, written from lecture content*

**39.** Which best explains why a lesion of one medial longitudinal fasciculus impairs ADDUCTION of the ipsilateral eye specifically, rather than some other eye movement?

- A. Because the MLF specifically carries the signal from the contralateral abducens nucleus to the ipsilateral oculomotor nucleus's medial rectus subnucleus, which is what drives eye adduction during conjugate horizontal gaze
- B. Because the MLF has no relationship to any specific eye muscle
- C. Because the MLF controls only vertical eye movements
- D. Because the MLF is part of the visual (not motor) pathway
- E. Because adduction is controlled by the trochlear nerve, unrelated to the MLF

*† original question, written from lecture content*

**40.** Which region does the Ascending Reticular Activating System (ARAS) NOT directly project to?

- A. Thalamus (intralaminar nuclei)
- B. Basal forebrain
- C. Hypothalamus
- D. Cerebral cortex, diffusely (via thalamic relay)
- E. The cerebellar deep nuclei as its primary/defining target

**41.** What is the general term for the diffuse network of interconnected neurons running through the core of the brainstem, from the medulla to the midbrain?

- A. Reticular formation
- B. Corticospinal tract
- C. Medial lemniscus
- D. Cerebellar peduncle
- E. Corpus callosum

*† original question, written from lecture content*

## Lecture 11 — Special Sense Organs & Its Development (8 questions)

**42.** Cochlear hair cells are innervated by bipolar neurons whose cell bodies reside in which structure?

- A. Geniculate ganglion
- B. Spiral ganglion (within the modiolus of the cochlea)
- C. Vestibular ganglion
- D. Petrosal ganglion
- E. Dorsal root ganglion

*† original question, written from lecture content*



**43.** Inner hair cells make up what proportion of the afferent fibers of the cochlear nerve, despite being far fewer in number than outer hair cells?

- A. About 5%
- B. About 50%
- C. About 95%
- D. 0%, they have no afferent innervation
- E. Exactly equal to outer hair cells

*† original question, written from lecture content*

**44.** Stereocilia of vestibular/cochlear hair cells are anchored where, and how do they relate to the kinocilium in mechanotransduction?

- A. They are embedded in an overlying membrane (e.g., tectorial membrane in the cochlea) and their deflection relative to the kinocilium determines depolarization vs. hyperpolarization
- B. They have no relationship to any overlying structure
- C. They are located inside the cell body, not on the apical surface
- D. They always cause hyperpolarization regardless of direction
- E. Kinocilium is absent in all mature mammalian hair cells

*† original question, written from lecture content*

**45.** The three semicircular canals detect which type of head movement?

- A. Linear acceleration only (e.g., gravity, forward/backward motion)
- B. Angular (rotational) acceleration, each canal oriented in a different plane
- C. Sound frequency only
- D. Static head position only, with no dynamic component
- E. Taste changes

*† original question, written from lecture content*

**46.** Which histological feature distinguishes the utricle and saccule (otolith organs) from the semicircular canal ampullae?

- A. The maculae of the utricle/saccule have an otolithic membrane embedded with calcium carbonate crystals (otoconia), adding inertial mass; the ampullary crista instead has a gelatinous cupula with no crystals
- B. They have identical histology with no differences
- C. Only the ampullae contain hair cells
- D. Only the maculae contain hair cells
- E. The utricle has no sensory epithelium at all

*† original question, written from lecture content*

**47.** Which structure supports the organ of Corti and is set into vibration by traveling sound waves, with different regions tuned to different frequencies (tonotopic organization)?

- A. Tectorial membrane
- B. Basilar membrane
- C. Reissner's membrane
- D. Stria vascularis
- E. Round window membrane

*† original question, written from lecture content*

**48.** Presbycusis (age-related hearing loss) classically affects which frequency range first, and which cochlear structure is most vulnerable?

- A. Low frequencies first, affecting the cochlear apex
- B. High frequencies first, affecting outer hair cells at the cochlear base
- C. All frequencies equally, with no particular pattern
- D. Only affects middle ear ossicles, not the cochlea
- E. Only affects the vestibular system, not hearing

*† original question, written from lecture content*

**49.** Which structure transmits sound vibrations from the tympanic membrane to the oval window, providing mechanical impedance matching between air and cochlear fluid?

- A. The three ossicles (malleus, incus, stapes) acting as a lever system
- B. The Eustachian tube
- C. The round window
- D. The semicircular canals
- E. The vestibule alone

*† original question, written from lecture content*

### Lectures 12–13 — Somatosensory Pathways / Taste Sensation (8 questions)

**50.** Which structure is responsible for relaying UNCONSCIOUS proprioceptive information from the lower limb (via the dorsal spinocerebellar tract) toward the cerebellum, synapsing first at Clarke's nucleus (C8-L2/3)?

- A. Dorsal root ganglion neurons directly to the cerebellum, with no relay
- B. First-order neuron → Clarke's nucleus (second-order neuron) → dorsal spinocerebellar tract → cerebellum
- C. Thalamus → cerebellum
- D. Directly via the corticospinal tract
- E. Via the spinothalamic tract

**51.** What structure is responsible for unconscious proprioceptive input from the SACRAL spinal cord levels (S2-S4), given that Clarke's nucleus does not extend this far caudally?

- A. Direct dorsal column ascent with no relay
- B. Sacral proprioceptive afferents ascend and effectively synapse in the region of Clarke's nucleus equivalent higher up, or use the ventral spinocerebellar pathway
- C. They have no route to the cerebellum at all
- D. They travel exclusively via the spinothalamic tract
- E. They bypass the spinal cord entirely

**52.** Which pathway carries CONSCIOUS proprioception, fine touch, and vibration sense, as opposed to the unconscious proprioceptive spinocerebellar pathways?

- A. Lateral spinothalamic tract
- B. Dorsal column-medial lemniscus (DCML) pathway
- C. Dorsal spinocerebellar tract
- D. Ventral spinocerebellar tract
- E. Anterior spinothalamic tract

*† original question, written from lecture content*

**53.** A lesion of the dorsal columns bilaterally (e.g., from vitamin B12 deficiency, subacute combined degeneration) would be expected to cause which finding?

- A. Loss of pain sensation with normal proprioception
- B. Sensory ataxia with a positive Romberg sign, loss of vibration and proprioception, normal pain/temperature
- C. Isolated motor weakness with no sensory loss
- D. Isolated loss of hearing
- E. Isolated Horner syndrome

*† original question, written from lecture content*

**54.** Which spinal cord tract carries fine touch, vibration, and conscious proprioception from the LOWER limb specifically, ascending in the most medial part of the dorsal columns?

- A. Fasciculus cuneatus (upper limb)
- B. Fasciculus gracilis (lower limb, medial dorsal column)
- C. Lateral spinothalamic tract
- D. Anterior spinothalamic tract
- E. Rubrospinal tract

*† original question, written from lecture content*

**55.** An unpleasant or distorted taste sensation is medically termed:

- A. Dysphagia
- B. Dysphonia
- C. Dysgeusia
- D. Dysarthria
- E. Dysmetria

**56.** When stereocilia of a taste/hair cell bend TOWARD the kinocilium, what is the physiological result?

- A. Hyperpolarization occurs
- B. K<sup>+</sup> channels close permanently
- C. Depolarization occurs
- D. Ca<sup>2+</sup> efflux increases
- E. Action potential frequency decreases

**57.** In the Gate-Control theory of pain, activated inhibitory spinal interneurons release enkephalins that act by:

- A. Blocking Na<sup>+</sup> channels presynaptically
- B. Opening presynaptic Ca<sup>2+</sup> channels
- C. Inhibiting presynaptic Ca<sup>2+</sup> influx and opening postsynaptic K<sup>+</sup> channels, reducing substance P release and dampening pain transmission
- D. Stimulating NMDA receptors
- E. Increasing glutamate release

## Lecture 14 — Sensory Receptor Transduction & Neurotransmitters (10 questions)

**58.** What is the term for the time delay between an action potential arriving at a presynaptic terminal and its effect on the postsynaptic membrane?

- A. Synaptic delay
- B. Synaptic fatigue

59. Which of the following receptors is ionotropic (directly gates an ion channel upon ligand binding)?
- A. AMPA receptor
  - B. mGluR6
  - C. GABA-B receptor
  - D. 5-HT1 receptor
60. What is the mechanism of fluoxetine?
- A. Selective serotonin reuptake inhibitor (blocks SERT)
  - B. Blocks D2 dopamine receptor
  - C. Blocks peripheral and central 5-HT3 receptors
61. What is the mechanism of the antiemetic drug ondansetron?
- A. Selective serotonin reuptake inhibitor
  - B. Blocks D2 receptor
  - C. Blocks peripheral and central 5-HT3 receptors
62. What is the mechanism of haloperidol?
- A. Selective serotonin reuptake inhibitor
  - B. Blocks D2 dopamine receptor
  - C. Blocks peripheral and central 5-HT3 receptors
63. A patient is splashed with hot cooking oil. Which channel is the primary receptor for NOXIOUS heat in this scenario?
- A. TRPV2 (very high heat threshold, extreme noxious heat)
  - B. TRPV4
  - C. TRPM8 (cold receptor — wrong category entirely)
64. A woman presents with fatigue, social withdrawal, and depressed mood. Which neurotransmitter system is most likely implicated?
- A. Dopamine
  - B. Glutamate
  - C. Serotonin
  - D. Acetylcholine
  - E. Epinephrine
65. Anosmia and ageusia (loss of smell and taste) can result from downregulation of which receptor, also notable for its role as the entry receptor for SARS-CoV-2?
- A. Dopamine D2 receptor
  - B. Muscarinic acetylcholine receptor
  - C. ACE2 (angiotensin-converting enzyme 2)
  - D. 5-HT3 receptor
  - E. NMDA receptor
66. Which neurotransmitter-transporter pairing is correct?
- A. VGAT - GABA (correct vesicular pairing)
  - B. SERT - Dopamine
  - C. DAT - Acetylcholine
  - D. EAATs - Norepinephrine
  - E. Glutaminase - Glutamine

**67.** A patient overdoses on a drug that directly opens GABA-A chloride channels even in the complete absence of GABA, causing profound, potentially fatal respiratory depression. Which drug class does this describe, and why is it more dangerous than benzodiazepines in overdose?

- A. Benzodiazepines, because they always require GABA to be present
- B. Barbiturates, because at high doses they can directly gate the Cl<sup>-</sup> channel independent of GABA, producing unopposed, potentially fatal CNS/respiratory depression
- C. SSRIs, which have no GABA-A activity at all
- D. Opioids, which act on a completely unrelated receptor system with no ceiling effect described here
- E. Antipsychotics, which block dopamine receptors, not GABA-A

*† original question, written from lecture content*

## Lectures 15–16 — Visual System / Auditory & Vestibular Systems (10 questions)

**68.** Which correctly arranges retinal layers from the rod/cone (photoreceptor) layer toward the optic nerve fiber layer?

- A. Rods/cones → Inner nuclear → Outer nuclear → Ganglion cell → Optic nerve fiber
- B. Rods/cones → Outer nuclear → Inner plexiform → Inner nuclear → Optic nerve fiber
- C. Rods/cones → Outer nuclear → Inner nuclear → Ganglion cell → Optic nerve fiber
- D. Rods/cones → Ganglion cell → Inner nuclear → Optic nerve fiber

**69.** A patient has a left pupil that does not constrict and is larger than the right, with associated ptosis and 'down and out' eye deviation. Where is the lesion?

- A. Optic tract
- B. Optic nerve
- C. Optic chiasm
- D. CN III (oculomotor nerve)
- E. Optic radiation

**70.** Surface ectoderm gives rise to which eye structure?

- A. Lens
- B. Iris
- C. Corneal endothelium
- D. Fibrous tunic (sclera) connective tissue
- E. Retina

**71.** Which specialization of the fovea centralis increases visual acuity there?

- A. Abundant blood vessels overlying it
- B. A high ratio of photoreceptor cells to ganglion cells
- C. Lateral displacement of overlying ganglion/inner retinal cell layers, minimizing light scatter before reaching photoreceptors
- D. Increased density of rod cells

**72.** What structure is damaged to produce a RIGHT homonymous hemianopia?

- A. Left optic tract (or fibers further downstream on the left visual pathway)
- B. Optic chiasm
- C. Superior left lateral geniculate body region alone
- D. Inferior left lateral geniculate body region alone

- 73.** What structure detects vertical linear acceleration, as when an elevator starts moving up or down?
- A. Superior semicircular canal
  - B. Lateral (horizontal) semicircular canal
  - C. Posterior semicircular canal
  - D. Saccule (oriented to detect vertical linear acceleration)
  - E. Utricle
- 74.** What structure primarily detects tilting the head UP (vertical linear head position change)?
- A. Utricle
  - B. Saccule
  - C. Horizontal semicircular canal
  - D. Superior semicircular canal
  - E. Cochlea
- 75.** What structure is primarily activated by nodding the head forward (in the sagittal/vertical rotational plane)?
- A. Utricle
  - B. Saccule
  - C. Horizontal semicircular canal
  - D. Superior (anterior) semicircular canal
  - E. Cochlea
- 76.** When you extend your head backward (posterior tilt), which vestibular receptor is primarily stimulated?
- A. Superior semicircular canal
  - B. Lateral semicircular canal
  - C. Posterior semicircular canal
  - D. Saccule
  - E. Utricle
- 77.** Where is the vomiting center located?
- A. Vestibular nuclei
  - B. Nucleus ambiguus
  - C. Area postrema
  - D. Superior and inferior salivatory nuclei
  - E. Nucleus solitarius

# Answer Key & Explanations

## Lectures 1–16 · 77 Questions

### Lecture 1 — Introduction & Organization of the CNS (5 questions)

#### Q1. Which statement about gray and white matter distribution is correct?

✓ **Correct: B. In the cerebrum, gray matter forms the cortex and white matter lies deep to it**

Cerebral/cerebellar cortex = gray matter (cell bodies) on the outside; the underlying white matter is myelinated axon tracts connecting cortical areas.

✗ **A is wrong:** *This is backwards for the spinal cord — the spinal cord actually has gray matter centrally (the classic 'butterfly') surrounded by white matter tracts peripherally, the opposite arrangement to the cerebrum.*

*TIP: Spinal cord and cerebrum have OPPOSITE gray/white arrangements — a favorite flip-the-diagram trick question.*

#### Q2. A newborn is found to have a portion of spinal cord and meninges protruding through a vertebral defect, covered only by a thin membrane....

✓ **Correct: C. Myelomeningocele**

Myelomeningocele = neural tissue (spinal cord/roots) PLUS meninges herniate through the vertebral defect — the most severe open form of spina bifida.

✗ **B is wrong:** *A meningocele herniates meninges and CSF only — no neural tissue — so it usually has milder or no neurological deficit.*

✗ **A is wrong:** *Spina bifida occulta is the mildest form: just a bony vertebral arch defect with skin intact over it, often an incidental finding.*

*TIP: Rank spina bifida severity: occulta (bone only, hidden) < meningocele (meninges herniate) < myelomeningocele (meninges + cord herniate, worst).*

#### Q3. Which cranial nerve numbers correspond to nerves that are purely sensory?

✓ **Correct: B. I, II, VIII**

CN I (olfactory), CN II (optic), and CN VIII (vestibulocochlear) carry only special sensory information — no motor fibers.

✗ **A is wrong:** *III, IV, and VI are the purely MOTOR nerves (oculomotor, trochlear, abducens) — the opposite category from what's being asked.*

*TIP: Group the 12 CNs into 3s: Sensory only (I, II, VIII), Motor only (III, IV, VI, XI, XII), Mixed (V, VII, IX, X).*

#### Q4. Dorsal root ganglia contain the cell bodies of which type of neuron?

✓ **Correct: C. First-order (primary) sensory neurons**

DRG house the pseudounipolar cell bodies of primary afferent (first-order sensory) neurons carrying input from the periphery into the spinal cord.

✗ **A is wrong:** *Lower motor neuron cell bodies live in the ventral horn of the spinal cord gray matter, not in a peripheral ganglion.*

#### Q5. A patient has a lesion that damages both anterior and posterior spinal nerve roots equally at one level. Which functional deficit...

✓ **Correct: C. Combined sensory and motor loss at that segmental level**

The posterior (dorsal) root is sensory and the anterior (ventral) root is motor; damaging both together at a spinal nerve level knocks out both modalities in that dermatome/myotome.

✗ **A is wrong:** *This only accounts for the dorsal root — ignoring the ventral (motor) root damage described in the stem.*

*TIP: Dorsal = sensory (Dorsal = Down = sensory comes down from the periphery... or just remember 'Bell-Magendie law': dorsal root sensory, ventral root motor.*

## Lecture 2 — Neurohistology (Neuron & Neuroglia) (3 questions)

**Q6. If the tract carrying FAST, sharp, well-localized pain is damaged, which fiber type is responsible for that lost sensation?**

✓ **Correct: D. Neo-spinothalamic tract (A-δ) fibers**

Fast, sharp, well-localized ('first') pain is carried by thinly myelinated A-δ fibers via the neospinothalamic tract, which projects rather directly to VPL and then somatosensory cortex — hence good localization.

✗ *B is wrong: C fibers carry SLOW, dull, poorly-localized ('second') pain via the paleospinothalamic tract — the opposite pain quality from what's described.*

*TIP: Fast/sharp/localized = A-δ = neospinothalamic. Slow/dull/burning/poorly localized = C fiber = paleospinothalamic (more diffuse projections including to reticular formation/limbic system, which is why it's harder to pinpoint and more 'emotionally' unpleasant).*

**Q7. Which of the following is NOT a component of the blood-brain barrier?**

✓ **Correct: A. Oligodendrocyte**

The BBB's neurovascular unit is built from capillary endothelial cells (tight junctions), the basement membrane, pericytes, and astrocytic end-feet — oligodendrocytes play no structural role in it.

✗ *C is wrong: Pericytes are a genuine, often-forgotten component that helps regulate BBB permeability and capillary blood flow — students frequently (wrongly) exclude this one instead.*

*TIP: BBB team: Endothelium (tight junctions, the main seal) + Basement membrane + Pericytes + Astrocyte end-feet. Oligodendrocytes are not invited to this party — they're busy myelinating.*

**Q8. Which cell type is the resident immune cell of the CNS, analogous to macrophages elsewhere in the body?**

✓ **Correct: A. Microglia**

Microglia are mesoderm/yolk-sac-derived resident CNS immune cells that phagocytose debris and pathogens and mediate neuroinflammation.

✗ *D is wrong: Astrocytes support the BBB, metabolism, and repair (glial scarring) but are not the primary phagocytic immune surveillance cell — that's specifically microglia's job.*

## Lecture 3 — Properties of Neuron: RMP, Action Potential, Conduction (5 questions)

**Q9. Which of the following correctly describes when Na<sup>+</sup>/K<sup>+</sup>-ATPase activity increases the most?**

✓ **Correct: A. During periods of high neuronal firing, to restore ionic gradients dissipated by Na<sup>+</sup> influx/K<sup>+</sup> efflux**

Every action potential lets Na<sup>+</sup> in and K<sup>+</sup> out; the pump has to work harder (using more ATP) proportional to how much firing has occurred, to restore the resting gradients.

**Q10. When depolarizations from multiple simultaneous synaptic inputs on different parts of a neuron add together, this is called:**

✓ **Correct: B. Spatial summation**

Spatial summation = combining inputs arriving at different locations on the neuron at roughly the same time; temporal summation is combining repeated inputs at the SAME synapse arriving in rapid succession.

✗ *A is wrong: Temporal summation is the time-based version (same synapse, rapid-fire inputs) — the question specifically describes different synapses/locations, which is spatial.*

*TIP: SPATIAL = different SPACE (different synapses) at the same time. TEMPORAL = same synapse, different TIME (rapid repeats).*



**Q11. All of the following are true of chemical synapses EXCEPT:**

✓ **Correct: D. Transmission is instantaneous with essentially zero delay**

Chemical synapses always have a measurable synaptic delay (vesicle fusion, diffusion, receptor binding take real time) — instantaneous, zero-delay transmission describes ELECTRICAL synapses (gap junctions), not chemical ones.

*TIP: Electrical synapses (gap junctions) = fast, bidirectional, no delay, no amplification, but also no plasticity.  
Chemical synapses = slower, unidirectional, delayed, but capable of amplification and plasticity (learning/memory).  
Know both columns of this comparison table.*

**Q12. Which statement about graded potentials is INCORRECT?**

✓ **Correct: D. They obey the all-or-none law like action potentials**

Graded potentials are explicitly NOT all-or-none — their amplitude scales with stimulus intensity. The all-or-none law is the signature property of the action potential, the opposite category.

**Q13. During the absolute refractory period, why can no new action potential be generated no matter how strong the stimulus?**

✓ **Correct: B. Voltage-gated Na<sup>+</sup> channels are inactivated and cannot reopen until the membrane repolarizes**

Na<sup>+</sup> channels have an inactivation gate that closes shortly after opening and won't reset (become available again) until the membrane has repolarized back toward resting potential — no stimulus, however strong, can force them open during this state.

✗ **C is wrong:** That describes the RELATIVE refractory period (a stronger-than-normal stimulus CAN trigger an AP), not the absolute refractory period where triggering a new AP is flatly impossible.

*TIP: Absolute refractory = impossible, period (Na<sup>+</sup> channels inactivated). Relative refractory = possible but needs a bigger stimulus (partially hyperpolarized + some channels still inactivated).*

## Lecture 4 — Metabolic Pathways of Neurons & Nervous Tissue (5 questions)

**Q14. Why can hyperammonemia impair neuronal function even though ammonia detoxification happens mainly in astrocytes rather than neurons?**

✓ **Correct: A. Because astrocyte dysfunction/swelling disrupts the glutamate-glutamine cycle and lactate shuttle that neurons depend on for neurotransmitter recycling and fuel**

Neurons and astrocytes are metabolically interdependent (lactate shuttle, glutamate-glutamine cycle); when astrocytes are overwhelmed and swollen from ammonia detoxification, this whole support system for neurons breaks down, indirectly impairing neuronal neurotransmission and energy supply even though neurons themselves aren't doing the detoxifying.

*TIP: Big-picture takeaway questions like this ('why does X affect Y indirectly') are how a prof tests whether you understand the SYSTEM, not just isolated facts. Always ask yourself 'what depends on what?' when reviewing a pathway.*

**Q15. Which transporter class is primarily responsible for astrocytic uptake of the lactate that astrocytes then export to neurons?**

✓ **Correct: C. Monocarboxylate transporters (MCT1/4)**

Monocarboxylate transporters (MCT1/4 on astrocytes, MCT2 on neurons) move lactate (and pyruvate/ketone bodies) across cell membranes — the physical conduit of the astrocyte-neuron lactate shuttle.

✗ **A is wrong:** GLUT3 is a neuronal GLUCOSE transporter, not a lactate transporter — glucose and lactate use entirely different transporter families.

**Q16. Which glucose transporter is characteristic of astrocytes, versus the one preferentially expressed in neurons?**

✓ **Correct: A. Astrocyte: GLUT1; Neuron: GLUT3**

Astrocytes preferentially express GLUT1 (lower affinity, matches their role as the glucose-sensing/glycolytic 'first responder'); neurons preferentially express GLUT3 (higher affinity, ensures neurons can still pull in glucose even at low extracellular concentrations).

*TIP: Higher-affinity transporter goes to the cell type that absolutely cannot afford to run out of substrate — neurons (GLUT3) win the 'first dibs on glucose' contest over astrocytes (GLUT1).*

**Q17. The pentose phosphate pathway (PPP), activated in neurons especially under high glucose availability, primarily generates which product...**

✓ **Correct: B. NADPH, which regenerates reduced glutathione (GSH) to neutralize reactive oxygen species**

The PPP's main output relevant here is NADPH, the essential reducing cofactor for regenerating glutathione (GSSG → GSH) and recycling ascorbate — both central to neutralizing oxidative stress, to which neurons are especially vulnerable given their high metabolic rate.

**Q18. Which micronutrient deficiency is classically associated with Wernicke-Korsakoff syndrome, disrupting neuronal energy metabolism?**

✓ **Correct: A. Vitamin B1 (thiamine)**

Thiamine is a required cofactor for pyruvate dehydrogenase and  $\alpha$ -ketoglutarate dehydrogenase (both key TCA cycle enzymes); its deficiency (classically in chronic alcoholism) starves neurons of ATP production and causes the classic triad of confusion, ataxia, and ophthalmoplegia.

*TIP: Thiamine deficiency = TCA cycle bottleneck (can't run pyruvate/ $\alpha$ -KG dehydrogenase efficiently) = ATP crisis in the most metabolically-demanding brain regions first (mammillary bodies, thalamus) — that's why Wernicke's classically involves those structures.*

## Lectures 5–6 — Embryonic Development of the Nervous System / Development of the Eye (5 questions)

**Q19. Which embryonic structure gives rise to the pineal gland?**

✓ **Correct: A. Roof of the diencephalon (epithalamus)**

The pineal gland develops as an evagination of the diencephalic roof plate (epithalamus) — consistent with it being a genuine CNS/neuroectodermal structure.

**Q20. Which functional component is characteristic of neurons in the hypoglossal nucleus, located at the level of the myelencephalon (medulla)?**

✓ **Correct: E. GSE (general somatic efferent)**

The hypoglossal nerve (CN XII) is a pure motor nerve to tongue musculature (derived from occipital somites) — classified functionally as GSE, the same category as the somatic motor nuclei of III, IV, VI.

*✗ D is wrong: SVE (special visceral efferent) is the category for branchiomeric muscle motor nuclei (e.g., facial, trigeminal motor, nucleus ambiguus) — tongue muscles are NOT branchial arch derivatives, so hypoglossal is GSE, not SVE.*

*TIP: CN XII (hypoglossal) and CN III/IV/VI (eye movers) are the only pure GSE cranial nerve nuclei — because tongue and extraocular muscles both derive from somites/somitomeres, not pharyngeal arches.*

**Q21. Which structure does the optic cup differentiate into?**

✓ **Correct: C. Retina (both neural and pigmented layers) and epithelium of the iris/ciliary body**

The two layers of the optic cup become the neural retina (inner layer) and retinal pigment epithelium (outer layer), with the cup's rim extending forward to form the epithelial layers of the iris and ciliary body.

**Q22. Failure of the choroid (retinal) fissure to fuse normally results in which congenital anomaly?**

✓ **Correct: A. Coloboma (a gap/cleft in the iris, retina, or optic disc)**

A coloboma is a structural defect (notch/gap, classically in the inferior iris/retina/choroid/optic nerve) caused by incomplete fusion of the edges of the choroid fissure during the 5th-7th weeks.

*TIP: Colobomas are classically INFERIOR because the choroid fissure runs along the ventral/inferior aspect of the developing optic cup/stalk — if you see 'notch in the iris/retina,' think coloboma, and remember its typical location.*

**Q23. The spinal trigeminal nucleus (carrying facial pain/temperature) is derived from which embryonic structure?**

✓ **Correct: B. Alar plate of the developing brainstem (sensory column)**

Sensory brainstem nuclei (including the spinal trigeminal nucleus) derive from the alar plate — the dorsal, sensory-associated portion of the developing neural tube; motor nuclei derive from the ventral basal plate.

✗ **C is wrong:** The basal plate gives rise to **MOTOR** nuclei — the sulcus limitans is the developmental landmark separating dorsal alar (sensory) from ventral basal (motor), and this pattern holds throughout the brainstem and spinal cord.

*TIP: Alar = Afferent/sensory (dorsal). Basal = Basically motor (ventral). This alar/basal split, established early in neural tube development, explains the entire dorsal-sensory/ventral-motor organization you see throughout the spinal cord and brainstem.*

## Lecture 7 — Spinal Cord & Blood Supply (5 questions)

**Q24. A 53-year-old man is hospitalized for acute spinal cord infarction (anterior spinal artery territory). Which sensory modality is...**

✓ **Correct: C. Vibration and proprioception (dorsal column)**

Anterior spinal artery syndrome spares the dorsal columns (supplied instead by the paired posterior spinal arteries), producing a classic dissociated pattern: loss of motor function, pain, and temperature, with **PRESERVED** vibration/proprioception.

*TIP: Anterior spinal artery syndrome = motor + pain/temp LOST, proprioception/vibration SPARED (dorsal columns have their own separate posterior spinal artery supply). This is the mirror image of the posterior spinal artery syndrome pattern.*

**Q25. Which of the following is NOT part of the central nervous system?**

✓ **Correct: C. Dorsal root ganglion**

The dorsal root ganglion sits just outside the spinal cord (in or near the intervertebral foramen) and is a PNS structure housing primary sensory neuron cell bodies.

**Q26. If the anterior spinal artery is occluded, which structure/function is spared, consistent with its blood supply territory?**

✓ **Correct: C. Dorsal column (vibration/proprioception) function, supplied by the posterior spinal arteries instead**

Same key fact restated from the syndrome above: the dorsal columns are spared because they're supplied by the posterior spinal arteries, a separate vascular territory from the anterior spinal artery.

**Q27. Which finding is a primary MOTOR symptom of a complete spinal cord lesion (transection) below the level of injury?**

✓ **Correct: B. Spastic paralysis (initially flaccid in spinal shock, then spastic) below the lesion**

Complete cord transection interrupts descending corticospinal (and other motor) tracts, producing an acute phase of flaccid paralysis/areflexia ('spinal shock') that evolves over days-weeks into chronic spastic paralysis with hyperreflexia below the lesion, once spinal reflex circuits recover from the initial shock.

*TIP: Don't forget the TIME COURSE: immediately after transection = FLACCID (spinal shock), then LATER = SPASTIC. A question describing the acute setting expects flaccid; one describing weeks later expects spastic — a classic timing trap.*

**Q28. Which structure extends from the pia mater to the dura mater along the lateral aspect of the spinal cord, anchoring it and separating...**

✓ **Correct: A. Denticulate ligament**

The denticulate ligaments are lateral, tooth-like pial extensions that pierce the arachnoid to attach to the dura, stabilizing the spinal cord within the subarachnoid space and providing a useful surgical landmark between dorsal and ventral roots.

✗ **B is wrong:** The filum terminale anchors the CAUDAL tip of the spinal cord/conus medullaris to the coccyx — a longitudinal, midline structure, not the lateral tooth-like structures separating the roots.

## Lecture 8 — PNS: Cranial & Spinal Nerves, Nerve Plexuses (3 questions)

**Q29. Which functional component is NOT found in the facial nerve (CN VII)?**

✓ **Correct: E. GVA (general visceral afferent from thoracic/abdominal viscera)**

The facial nerve is a genuinely mixed nerve with SVE, GVE, SVA, and a small GSA component (external ear) — but it does NOT carry GVA fibers from thoracic/abdominal viscera; that's characteristic of the vagus nerve instead.

*TIP: CN VII is famous for carrying FOUR different functional components — a favorite 'which one does it NOT have' trick question, since students often assume a nerve either has one component or 'all of them.'*

**Q30. Which spinal nerves are NOT included in a somatic nerve plexus?**

✓ **Correct: B. T2-T12 (thoracic nerves, which mostly run as individual intercostal nerves)**

Most thoracic spinal nerves (roughly T2-T12) do NOT form a plexus — they run individually as intercostal (and subcostal) nerves, unlike the cervical, brachial, and lumbosacral levels which do interweave into plexuses.

**Q31. A 41-year-old woman has recurring episodes of sharp, electric-shock-like facial pain triggered by chewing and touching her cheek. Which...**

✓ **Correct: B. Trigeminal nerve (trigeminal neuralgia)**

This is the classic presentation of trigeminal neuralgia — brief, severe, electric-shock-like facial pain in a trigeminal (V2/V3 most commonly) distribution, often triggered by light touch, chewing, or talking.

## Lectures 9–10 — Brainstem: External & Internal Structures / Reticular Formation (10 questions)

**Q32. Which tract's second-order neurons cross the midline at the level of the medulla oblongata (rather than immediately upon entering the...**

✓ **Correct: B. Dorsal column-medial lemniscus pathway (crosses as internal arcuate fibers in the caudal medulla)**

Dorsal column axons ascend ipsilaterally all the way up to the caudal medulla, synapse in the gracile/cuneate nuclei, and THEN the second-order neurons cross as internal arcuate fibers to form the contralateral medial lemniscus — a distinctly later crossing point than the spinothalamic tract's early spinal cord crossing.

*X A is wrong: Spinothalamic tract fibers cross much earlier and lower — within about 1-2 spinal segments of entering the cord via the anterior white commissure — not at the medulla.*

*TIP: This crossing-LEVEL difference (spinothalamic crosses low, in the cord; DCML crosses high, in the medulla) is EXACTLY why Brown-Séquard syndrome produces opposite-side patterns for these two pathways below a cord lesion, but why a medullary lesion can selectively hit DCML crossing fibers alone.*

**Q33. Damage to which major component of the Ascending Reticular Activating System (ARAS) most directly impairs consciousness?**

✓ **Correct: D. Pons and midbrain (rostral brainstem reticular formation)**

The ARAS core is the pontomesencephalic (pons + midbrain) reticular formation, which projects to the thalamus and cortex to maintain wakefulness — bilateral damage here (or to its thalamic/cortical targets) is a classic cause of coma.

**Q34. What is the major anatomical site whose bilateral damage produces coma via ARAS disruption?**

✓ **Correct: B. Pons and midbrain reticular formation**

Same core concept: bilateral pontomesencephalic reticular formation damage (or bilateral thalamic/diffuse cortical damage) is required to produce coma — UNILATERAL cortical damage alone typically does not cause coma given the bilateral/diffuse nature of ARAS projections.

*TIP: Coma requires either (1) bilateral brainstem ARAS damage, or (2) bilateral (diffuse) cortical/thalamic damage — a UNILATERAL hemispheric lesion alone, however large, usually does NOT cause coma unless it secondarily compresses the brainstem (e.g., via herniation).*

**Q35. A patient presents with altered consciousness, irregular breathing patterns, and decorticate/decerebrate posturing. This clinical...**

✓ **Correct: B. Brainstem (reticular formation) dysfunction, possibly from herniation or direct injury**

Altered consciousness plus abnormal respiratory patterns plus abnormal posturing together localize to brainstem/reticular formation dysfunction (often from rising ICP causing herniation) — respiratory pattern and posturing type can even help localize the LEVEL of brainstem compromise (e.g., decorticate = above red nucleus, decerebrate = below it, at/below midbrain).

*TIP: Decorticate (arms flexed, legs extended) = lesion ABOVE the red nucleus (better prognosis). Decerebrate (arms AND legs extended) = lesion AT OR BELOW the red nucleus, more caudal, worse prognosis. As the lesion progresses caudally, posturing gets worse — a useful serial-exam trend to recognize.*

**Q36. Which cranial nerve nuclei are found at the level of the pons (not the medulla or midbrain)?**

✓ **Correct: B. CN V, VI, VII, VIII**

CN V (trigeminal motor/main sensory), VI (abducens), VII (facial), and VIII (vestibulocochlear) nuclei are all located in the pons — a useful grouping for localizing pontine lesions by which nerves are affected.

*TIP: Cranial nerve nuclei by level: Midbrain = III, IV. Pons = V, VI, VII, VIII. Medulla = IX, X, XI, XII (plus V's spinal nucleus extends down here too). Memorizing this ladder lets you localize ANY brainstem lesion just from which cranial nerves are affected.*

**Q37. Which structure, if damaged bilaterally in the pons, classically produces 'locked-in syndrome' (preserved consciousness and vertical eye...**

✓ **Correct: A. Bilateral ventral pons (corticospinal and corticobulbar tracts), sparing the tegmentum/reticular formation**

Locked-in syndrome results from a lesion (classically basilar artery occlusion) confined to the VENTRAL pons, destroying descending motor pathways (corticospinal/corticobulbar) while SPARING the dorsal tegmentum where the ARAS and vertical gaze centers live — hence consciousness and vertical eye movement are preserved even though the patient cannot move or speak.

*TIP: This is a devastating and famous localization lesson: ventral pons = 'the highway out' (motor tracts passing through). Dorsal pons/tegmentum = 'the control room' (reticular formation, cranial nerve nuclei). Damage the highway but spare the control room, and you get a fully conscious, aware patient trapped in a paralyzed body — locked-in syndrome.*

**Q38. Why does a lesion of the medial longitudinal fasciculus (MLF) classically cause internuclear ophthalmoplegia (INO)?**

✓ **Correct: A. The MLF connects the abducens nucleus on one side to the contralateral oculomotor (medial rectus) subnucleus, coordinating conjugate horizontal gaze**

The MLF is the internuclear connection allowing the abducens nucleus (lateral gaze) to simultaneously signal the contralateral oculomotor nucleus's medial rectus subnucleus, so both eyes move together during horizontal conjugate gaze; damaging it (classically in multiple sclerosis) causes impaired adduction of the ipsilateral eye on attempted lateral gaze, with nystagmus in the abducting eye.

*TIP: INO is a favorite MS localization question: the affected side is named for which eye fails to ADDUCT (not the side of the primary MLF lesion's laterality convention some students misremember) — always work out the pathway from scratch rather than memorizing a shortcut, since the naming convention trips people up.*

**Q39. Which best explains why a lesion of one medial longitudinal fasciculus impairs ADDUCTION of the ipsilateral eye specifically, rather...**

✓ **Correct: A. Because the MLF specifically carries the signal from the contralateral abducens nucleus to the ipsilateral oculomotor nucleus's medial rectus subnucleus, which is what drives eye adduction during conjugate horizontal gaze**

The MLF is specifically the relay from the contralateral abducens nucleus (lateral gaze signal) to the ipsilateral oculomotor medial rectus subnucleus — since medial rectus is the adductor muscle, damaging this specific relay selectively impairs adduction on attempted conjugate gaze, sparing other eye movements that don't depend on this particular circuit.

**Q40. Which region does the Ascending Reticular Activating System (ARAS) NOT directly project to?**

✓ **Correct: E. The cerebellar deep nuclei as its primary/defining target**

The ARAS's defining projections are to the thalamus (intralaminar nuclei), basal forebrain, and hypothalamus, which then relay diffusely to the cortex to maintain arousal — the cerebellum is not its primary/defining target pathway.

**Q41. What is the general term for the diffuse network of interconnected neurons running through the core of the brainstem, from the medulla...**

✓ **Correct: A. Reticular formation**

The reticular formation is exactly this: a diffuse, interconnected neuronal network extending through the core of the brainstem from the medulla up through the midbrain, involved in arousal, autonomic regulation, and motor/reflex modulation.

## Lecture 11 — Special Sense Organs & Its Development (8 questions)

**Q42. Cochlear hair cells are innervated by bipolar neurons whose cell bodies reside in which structure?**

✓ **Correct: B. Spiral ganglion (within the modiolus of the cochlea)**

The spiral ganglion, coiled within the bony modiolus at the center of the cochlea, contains the bipolar first-order sensory neurons whose peripheral processes contact hair cells and whose central processes form the cochlear nerve.

✗ *C is wrong: The vestibular ganglion (Scarpa's ganglion) serves the vestibular (balance) hair cells of the semicircular canals/utricle/sacculle, a separate ganglion from the cochlear spiral ganglion — even though both eventually join CN VIII.*

**Q43. Inner hair cells make up what proportion of the afferent fibers of the cochlear nerve, despite being far fewer in number than outer hair...**

✓ **Correct: C. About 95%**

Despite inner hair cells being a minority of total hair cells, they receive about 95% of the afferent (sensory) innervation from the cochlear nerve — they are the true sensory transducers sending sound information to the brain; outer hair cells are mainly EFFERENT-controlled amplifiers (cochlear mechanical tuning), not primary sensory transmitters.

✗ *A is wrong: 5% is actually the OUTER hair cell share of afferent innervation — the numbers are commonly swapped in student memory since outer hair cells are more numerous overall but carry far less of the actual afferent signal.*

*TIP: Outer hair cells: more numerous, mostly EFFERENT-controlled amplifiers (cochlear amplifier function), only ~5% of afferent fibers. Inner hair cells: fewer in number, but ~95% of afferent fibers — they're the ones actually 'talking' to the brain about sound.*

**Q44. Stereocilia of vestibular/cochlear hair cells are anchored where, and how do they relate to the kinocilium in mechanotransduction?**

✓ **Correct: A. They are embedded in an overlying membrane (e.g., tectorial membrane in the cochlea) and their deflection relative to the kinocilium determines depolarization vs. hyperpolarization**

Stereocilia (graded in height, embedded in the tectorial membrane in the cochlea, or the otolithic/cupular membrane in the vestibular system) bend when the overlying membrane shifts; bending TOWARD the tallest stereocilia/kinocilium opens mechanically-gated K<sup>+</sup> channels causing depolarization, while bending AWAY closes them, causing hyperpolarization.

*TIP: Direction relative to kinocilium is everything: TOWARD kinocilium = depolarization = more neurotransmitter release = more afferent firing. AWAY from kinocilium = hyperpolarization = less firing. This single directional rule underlies hearing, balance (linear and angular acceleration detection), and is tested constantly.*



**Q45. The three semicircular canals detect which type of head movement?**

✓ **Correct: B. Angular (rotational) acceleration, each canal oriented in a different plane**

The three semicircular canals (anterior, posterior, lateral/horizontal), each roughly orthogonal to the others, detect ANGULAR (rotational) acceleration in their respective planes via endolymph movement deflecting the cupula.

✗ *A is wrong: Linear acceleration (including gravity/head tilt) is detected by the OTOLITH organs (utricle and saccule), not the semicircular canals — a fundamental and frequently tested vestibular distinction.*

*TIP: Semicircular canals = ANGULAR/rotational acceleration (spinning). Utricle + saccule (otolith organs) = LINEAR acceleration + static head position relative to gravity (tilting, elevator start/stop). Two different physical quantities, two different organ types.*

**Q46. Which histological feature distinguishes the utricle and saccule (otolith organs) from the semicircular canal ampullae?**

✓ **Correct: A. The maculae of the utricle/saccule have an otolithic membrane embedded with calcium carbonate crystals (otoconia), adding inertial mass; the ampullary crista instead has a gelatinous cupula with no crystals**

Otoconia (calcium carbonate crystals) embedded in the otolithic membrane give the maculae extra inertial mass, allowing them to lag behind during linear acceleration/gravity shifts and deflect the underlying hair cell stereocilia — the ampullary cupula has no such crystals, relying instead on endolymph flow during rotational movement.

**Q47. Which structure supports the organ of Corti and is set into vibration by traveling sound waves, with different regions tuned to...**

✓ **Correct: B. Basilar membrane**

The basilar membrane's mechanical properties vary along its length (narrow/stiff at the base = high frequencies; wide/flexible at the apex = low frequencies), creating tonotopic organization — the physical basis of frequency discrimination.

*TIP: Base of cochlea = narrow, stiff basilar membrane = HIGH frequency. Apex = wide, floppy basilar membrane = LOW frequency. This base-high/apex-low tonotopic map is the anatomical basis for frequency coding and gets tested via both anatomy and clinical (presbycusis pattern) questions.*

**Q48. Presbycusis (age-related hearing loss) classically affects which frequency range first, and which cochlear structure is most vulnerable?**

✓ **Correct: B. High frequencies first, affecting outer hair cells at the cochlear base**

Presbycusis classically begins with HIGH-frequency hearing loss, reflecting cumulative damage to outer (and eventually inner) hair cells at the cochlear BASE — the base being tuned to high frequencies and cumulatively exposed to the most mechanical stress over a lifetime.

**Q49. Which structure transmits sound vibrations from the tympanic membrane to the oval window, providing mechanical impedance matching...**

✓ **Correct: A. The three ossicles (malleus, incus, stapes) acting as a lever system**

The ossicular chain (malleus-incus-stapes) provides mechanical impedance matching (via lever action and the area ratio between the larger tympanic membrane and smaller oval window) — without this amplification, most sound energy would simply reflect off the fluid-air interface rather than being transmitted into the cochlea.

## Lectures 12–13 — Somatosensory Pathways / Taste Sensation (8 questions)

**Q50. Which structure is responsible for relaying UNCONSCIOUS proprioceptive information from the lower limb (via the dorsal spinocerebellar...**

✓ **Correct: B. First-order neuron → Clarke's nucleus (second-order neuron) → dorsal spinocerebellar tract → cerebellum**

Unconscious proprioceptive input from the lower limb synapses first onto second-order neurons in Clarke's nucleus (dorsal nucleus, present roughly C8-L2/3), which then form the DORSAL spinocerebellar tract, an uncrossed pathway carrying fine proprioceptive/movement information to the ipsilateral cerebellum for real-time motor coordination.

*TIP: Clarke's nucleus is present only C8-L2/3 — below L2/3, unconscious lower-limb proprioception instead uses an alternate route (spinal border cells → ventral spinocerebellar tract) since there's no Clarke's nucleus at those lower levels to synapse on directly. This nucleus-only-at-certain-levels fact is a favorite trick.*

**Q51. What structure is responsible for unconscious proprioceptive input from the SACRAL spinal cord levels (S2-S4), given that Clarke's...**

✓ **Correct: B. Sacral proprioceptive afferents ascend and effectively synapse in the region of Clarke's nucleus equivalent higher up, or use the ventral spinocerebellar pathway**

Since Clarke's nucleus proper doesn't extend into the sacral cord, unconscious proprioceptive signals from S2-S4 instead ascend and effectively relay via the ventral spinocerebellar tract pathway (using spinal border cells as the functional equivalent further down), ultimately still reaching the cerebellum for lower-limb coordination.

**Q52. Which pathway carries CONSCIOUS proprioception, fine touch, and vibration sense, as opposed to the unconscious proprioceptive...**

✓ **Correct: B. Dorsal column-medial lemniscus (DCML) pathway**

The DCML pathway is the one that reaches conscious awareness via the thalamus (VPL) and primary somatosensory cortex — the spinocerebellar tracts, by contrast, carry proprioceptive information that never reaches consciousness, feeding directly into cerebellar motor coordination circuits instead.

*TIP: Key distinction: DCML = CONSCIOUS proprioception (you can tell where your limb is). Spinocerebellar tracts = UNCONSCIOUS proprioception (used only by the cerebellum for automatic movement coordination, never reaches your awareness). Same general 'sense,' two totally different destinations and purposes.*

**Q53. A lesion of the dorsal columns bilaterally (e.g., from vitamin B12 deficiency, subacute combined degeneration) would be expected to...**

✓ **Correct: B. Sensory ataxia with a positive Romberg sign, loss of vibration and proprioception, normal pain/temperature**

Subacute combined degeneration classically affects the dorsal columns (and lateral corticospinal tracts) bilaterally, sparing the spinothalamic tracts — producing the classic dissociated pattern of proprioceptive/vibratory loss with sensory ataxia (positive Romberg), while pain and temperature sensation remain intact.

**Q54. Which spinal cord tract carries fine touch, vibration, and conscious proprioception from the LOWER limb specifically, ascending in the...**

✓ **Correct: B. Fasciculus gracilis (lower limb, medial dorsal column)**

The dorsal columns are somatotopically organized: fasciculus gracilis (medial) carries LOWER limb/lower trunk input, while fasciculus cuneatus (lateral, added above T6) carries UPPER limb/upper trunk input — a favorite localization detail.

✗ **A is wrong: Fasciculus cuneatus is the UPPER limb pathway, and it's positioned LATERAL (not medial) to the gracile fasciculus — the exact opposite pairing (nerve/position) from what this lower-limb question describes.**

*TIP: Dorsal column somatotopy: Gracile (medial, 'G for legs, Ground-level, lower/medial') = lower body. Cuneatus (lateral, added above T6) = upper body. This medial-leg/lateral-arm organization is preserved all the way up into the medial lemniscus too.*

**Q55. An unpleasant or distorted taste sensation is medically termed:**

✓ **Correct: C. Dysgeusia**

Dysgeusia = distorted/unpleasant taste. The other 'dys-' terms all refer to unrelated deficits: dysphagia (swallowing), dysphonia (voice), dysarthria (articulation), dysmetria (cerebellar movement accuracy).

*TIP: 'Dys-' word bank, don't mix these up: -phagia=swallowing, -phonia=voice/phonation, -geusia=taste, -arthria=speech articulation, -metria=cerebellar movement accuracy, -osmia=smell (anosmia=no smell). Five to seven similar-sounding terms is a classic examiner trap — build your own glossary.*

**Q56. When stereocilia of a taste/hair cell bend TOWARD the kinocilium, what is the physiological result?**

✓ **Correct: C. Depolarization occurs**

Bending toward the kinocilium opens mechanically-gated cation channels, causing depolarization and increased neurotransmitter release/afferent firing — the opposite of bending away, which hyperpolarizes the cell.



**Q57. In the Gate-Control theory of pain, activated inhibitory spinal interneurons release enkephalins that act by:**

✓ **Correct: C. Inhibiting presynaptic Ca<sup>2+</sup> influx and opening postsynaptic K<sup>+</sup> channels, reducing substance P release and dampening pain transmission**

Enkephalins act at opioid receptors to REDUCE presynaptic Ca<sup>2+</sup> influx (less neurotransmitter/substance P release from the primary nociceptive afferent) AND increase postsynaptic K<sup>+</sup> conductance (hyperpolarizing the second-order pain neuron) — a dual pre- and postsynaptic inhibitory mechanism dampening the ascending pain signal.

*TIP: Gate-control full circuit: large-diameter mechanoreceptor (A-β) input → activates inhibitory spinal interneurons → enkephalin release → presynaptic Ca<sup>2+</sup>↓ AND postsynaptic K<sup>+</sup>↑ → less substance P, less pain transmission. This is literally why rubbing an injury ('gating' pain with touch input) actually works physiologically.*

## Lecture 14 — Sensory Receptor Transduction & Neurotransmitters (10 questions)

**Q58. What is the term for the time delay between an action potential arriving at a presynaptic terminal and its effect on the postsynaptic...**

✓ **Correct: A. Synaptic delay**

Synaptic delay (~0.5-1ms) reflects the time needed for Ca<sup>2+</sup> entry, vesicle fusion, neurotransmitter diffusion, and receptor binding — the real, measurable time cost of chemical (as opposed to electrical) synaptic transmission.

✗ *B is wrong: Synaptic fatigue refers to a DECREASE in synaptic efficacy with sustained/repeated use (neurotransmitter depletion), a completely different phenomenon from the fixed time delay of a single transmission event.*

**Q59. Which of the following receptors is ionotropic (directly gates an ion channel upon ligand binding)?**

✓ **Correct: A. AMPA receptor**

AMPA receptors are ionotropic glutamate receptors (ligand-gated Na<sup>+</sup>/K<sup>+</sup> channels), providing fast excitatory transmission — the other three listed (mGluR6, GABA-B, 5-HT<sub>1</sub>) are all metabotropic (G-protein-coupled), acting through slower second-messenger cascades.

*TIP: Ionotropic = fast, direct ion channel (AMPA, NMDA, GABA-A, glycine, nicotinic ACh, 5-HT<sub>3</sub>). Metabotropic = slow, G-protein-coupled (mGluRs, GABA-B, muscarinic ACh, most 5-HT subtypes, dopamine, adrenergic). '3' is the magic ionotropic serotonin number to remember: 5-HT<sub>3</sub> is the ONE ionotropic serotonin receptor; all other 5-HT subtypes are metabotropic.*

**Q60. What is the mechanism of fluoxetine?**

✓ **Correct: A. Selective serotonin reuptake inhibitor (blocks SERT)**

Fluoxetine is a classic SSRI, blocking SERT to increase synaptic serotonin.

**Q61. What is the mechanism of the antiemetic drug ondansetron?**

✓ **Correct: C. Blocks peripheral and central 5-HT<sub>3</sub> receptors**

Ondansetron is a 5-HT<sub>3</sub> receptor antagonist, blocking serotonin's action at this specific ionotropic receptor subtype involved in the vomiting reflex.

**Q62. What is the mechanism of haloperidol?**

✓ **Correct: B. Blocks D<sub>2</sub> dopamine receptor**

Haloperidol is a typical (first-generation) antipsychotic, blocking D<sub>2</sub> dopamine receptors.

*TIP: Three drugs, three completely different single-target mechanisms tested back to back (fluoxetine=SERT, ondansetron=5-HT<sub>3</sub>, haloperidol=D<sub>2</sub>) — a favorite examiner trick is to present these consecutively and swap the mechanisms between them.*

**Q63. A patient is splashed with hot cooking oil. Which channel is the primary receptor for NOXIOUS heat in this scenario?**

✓ **Correct: A. TRPV2 (very high heat threshold, extreme noxious heat)**

For truly extreme/noxious heat (like hot oil, well above the ~43°C threshold of TRPV1), TRPV2 is the higher-threshold heat nociceptor channel recruited — distinct from TRPV1's more moderate noxious-heat range.

*TIP: Thermal TRP channel ladder by temperature: TRPM8 (cool, ~menthol) < TRPV4/TRPV3 (warm) < TRPV1 (hot/noxious, ~43°C+, capsaicin) < TRPV2 (extreme heat, >52°C). The hotter the stimulus, the higher up this ladder you go.*

**Q64. A woman presents with fatigue, social withdrawal, and depressed mood. Which neurotransmitter system is most likely implicated?**

✓ **Correct: C. Serotonin**

Same core depression-serotonin link tested again — a very frequently repeated exam fact worth over-learning.

**Q65. Anosmia and ageusia (loss of smell and taste) can result from downregulation of which receptor, also notable for its role as the entry...**

✓ **Correct: C. ACE2 (angiotensin-converting enzyme 2)**

ACE2 is expressed on olfactory support cells (sustentacular cells) and is both the receptor SARS-CoV-2 uses for cellular entry and a plausible mechanistic link to COVID-19-associated anosmia/ageusia — a great example of a basic science fact suddenly becoming extremely clinically relevant.

**Q66. Which neurotransmitter-transporter pairing is correct?**

✓ **Correct: A. VGAT - GABA (correct vesicular pairing)**

VGAT loads GABA into synaptic vesicles — a correctly matched pair. The other four options mismatch a transporter/enzyme with the wrong neurotransmitter (SERT is for serotonin not dopamine; DAT is for dopamine not ACh; EAATs are for glutamate not norepinephrine; glutaminase is an enzyme, not a transporter, and it converts glutamine TO glutamate, it doesn't transport glutamine per se).

**Q67. A patient overdoses on a drug that directly opens GABA-A chloride channels even in the complete absence of GABA, causing profound...**

✓ **Correct: B. Barbiturates, because at high doses they can directly gate the Cl<sup>-</sup> channel independent of GABA, producing unopposed, potentially fatal CNS/respiratory depression**

Barbiturates, unlike benzodiazepines, can directly open the GABA-A chloride channel even without GABA bound, at sufficiently high concentrations — this GABA-independent action removes any physiological 'ceiling' on inhibition, making barbiturate overdose far more likely to cause fatal respiratory arrest than benzodiazepine overdose (where the effect is capped by needing GABA to be present).

*TIP: This 'why is X more dangerous than Y' framing is a hallmark of a well-designed critical-thinking question — always be ready to explain not just the mechanism, but its clinical CONSEQUENCE (here: no ceiling effect = more dangerous in overdose).*

## Lectures 15–16 — Visual System / Auditory & Vestibular Systems (10 questions)

**Q68. Which correctly arranges retinal layers from the rod/cone (photoreceptor) layer toward the optic nerve fiber layer?**

✓ **Correct: C. Rods/cones → Outer nuclear → Inner nuclear → Ganglion cell → Optic nerve fiber**

Correct order from photoreceptor toward the vitreous/optic nerve: photoreceptor segments → outer nuclear layer (photoreceptor cell bodies) → outer plexiform → inner nuclear layer (bipolar/horizontal/amacrine cell bodies) → inner plexiform → ganglion cell layer → nerve fiber layer.

*TIP: Remember: light must pass THROUGH most retinal layers before hitting the photoreceptors (the retina is 'inside-out' relative to a camera) — but the SIGNAL then flows the opposite direction, from photoreceptor outward toward the ganglion cells and optic nerve, which is the direction this question tests.*

**Q69. A patient has a left pupil that does not constrict and is larger than the right, with associated ptosis and 'down and out' eye...**

✓ **Correct: D. CN III (oculomotor nerve)**

Dilated, fixed pupil + ptosis + down-and-out eye = complete CN III palsy — the parasympathetic pupillary fibers on the outside of the nerve are compressed, explaining the pupil involvement.

**Q70. Surface ectoderm gives rise to which eye structure?**

✓ **Correct: A. Lens**

The lens is the classic surface ectoderm derivative (via lens placode/vesicle) — the other options are neuroectoderm (iris epithelium, retina) or neural crest/mesoderm (corneal endothelium, sclera) derived.

**Q71. Which specialization of the fovea centralis increases visual acuity there?**

✓ **Correct: C. Lateral displacement of overlying ganglion/inner retinal cell layers, minimizing light scatter before reaching photoreceptors**

At the fovea, the inner retinal layers (ganglion cells, bipolar cells, etc.) are pushed laterally out of the way, letting light hit the densely-packed cone photoreceptors with minimal obstruction/scatter — a key structural adaptation for maximal visual acuity.

✗ *D is wrong: The fovea is actually CONE-dominated (essentially no rods at its very center) — increased ROD density is the opposite of what makes the fovea special; rods dominate the peripheral retina instead, favoring sensitivity over acuity.*

*TIP: Fovea = cones only, best acuity, worst in dim light (no rods = can't see faint stars if you look straight at them — that's why astronomers use averted vision, shifting the image to the rod-rich periphery).*

**Q72. What structure is damaged to produce a RIGHT homonymous hemianopia?**

✓ **Correct: A. Left optic tract (or fibers further downstream on the left visual pathway)**

A right homonymous hemianopia (loss of the right visual field in BOTH eyes) results from a lesion of the LEFT visual pathway AFTER the chiasm (left optic tract, radiation, or occipital cortex) — post-chiasmal lesions always cause a homonymous (same-side) field cut, on the side OPPOSITE the lesion.

*TIP: Golden rule: PRE-chiasmal lesions (optic nerve) = monocular defects. AT the chiasm = bitemporal (heteronymous). POST-chiasmal (tract, radiation, cortex) = homonymous, on the side of the visual field OPPOSITE to the lesion side. Always work out which 'zone' the lesion is in first.*

**Q73. What structure detects vertical linear acceleration, as when an elevator starts moving up or down?**

✓ **Correct: D. Saccule (oriented to detect vertical linear acceleration)**

The saccule's maculae are oriented in a roughly VERTICAL plane, making it the primary detector of vertical linear acceleration (elevator start/stop, jumping) — while the utricle's horizontally-oriented macula detects horizontal linear acceleration instead.

✗ *E is wrong: The utricle detects HORIZONTAL linear acceleration (like a car starting to move forward), the perpendicular plane from what this vertical-elevator scenario is testing.*

*TIP: Saccule = VERTICAL linear acceleration (elevator, jumping) — think 'Saccule = Straight up and down.' Utricle = HORIZONTAL linear acceleration (car accelerating forward) — think 'Utricle = yoU're driving forward.'*

**Q74. What structure primarily detects tilting the head UP (vertical linear head position change)?**

✓ **Correct: B. Saccule**

Same saccule/vertical-plane logic as the elevator question — tilting the head to change its vertical position relative to gravity is detected by the vertically-oriented saccular macula.

**Q75. What structure is primarily activated by nodding the head forward (in the sagittal/vertical rotational plane)?**

✓ **Correct: D. Superior (anterior) semicircular canal**

Nodding ('yes' motion) is a rotational movement in the sagittal plane, detected by the superior (anterior) semicircular canal, which is oriented to sense exactly this plane of angular acceleration.

*TIP: Semicircular canal orientation-to-movement mapping: Horizontal canal = shaking head 'no' (yaw). Anterior/superior canal = nodding 'yes' (pitch). Posterior canal = tilting head toward shoulder (roll). Each canal is tuned to detect rotation specifically in its own anatomical plane.*

**Q76. When you extend your head backward (posterior tilt), which vestibular receptor is primarily stimulated?**

✓ **Correct: A. Superior semicircular canal**

Extending the head backward is a rotational (pitch-plane) movement, sensed by the anterior/superior semicircular canal — the same canal responsible for the forward-nodding motion, since it detects rotation in that plane in EITHER direction.

**Q77. Where is the vomiting center located?**

✓ **Correct: C. Area postrema**

Area postrema (a circumventricular organ with a leaky BBB) functions as the chemoreceptor trigger zone that detects blood-borne emetogenic substances and directly triggers vomiting.

*TIP: Note the connection to vestibular input here too: motion sickness works because vestibular nuclei ALSO project to the vomiting center/area postrema — that's the whole physiological basis of why spinning/motion makes you nauseated.*